

6.12 Renting and Homeownership



Figure 6.26 Purchasing a home is a big investment, while renting is a lower cost alternative. (credit: "New construction, new development house for sale" by jongorey/houseandhammer.com, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Evaluate advantages and disadvantages of renting.
2. Evaluate advantages and disadvantages of homeownership.
3. Calculate the monthly payment for a mortgage and related interest cost.
4. Read and interpret an amortization schedule.
5. Solve application problems involving affordability of a mortgage.

After renting an apartment for 10 years, you realize that it may be time to purchase a home. Your job is stable, and you could use more space. It is time to investigate becoming a homeowner. What are the things that you must consider, and what is the financial benefit of owning as opposed to renting? This section is about the advantages, disadvantages, and costs of homeownership as opposed to renting.

Advantages and Disadvantages of Renting

When renting, you will likely sign a **lease**, which is a contract between a renter and a landlord. A **landlord** is the person or company that owns property that is rented. The lease will detail your responsibilities, restrictions on activities, deposits, fees, maintenance, repairs, and rent during the term of the lease. It also defines what your landlord can, and cannot, do with the property while you occupy the property.

Like leasing a car, there are advantages to renting but also some disadvantages. Some advantages are:

- Lower cost.
- Short-term commitment.
- Little to no maintenance cost. The landlord pays for or performs most maintenance.
- You need not stay at end of lease. Once the lease term is over (the lease is up), you are not obligated to stay.
- If renting in an apartment complex, there may be a pool, gym, or community room for renters to use.

Of course, there are disadvantage too:

- No tax incentives.
- Housing cost is not fixed. When the lease is up, the rent can change.
- No equity. When you are done living in a rental, you have built no value.
- Restrictions on occupants. There may be a limit on how many can live in the apartment.
- Restrictions on decorating. The property is not yours, so any decorating or improvements need landlord permission.
- Limits on pets. Permission for pets, and their number and type, will be set forth in the lease.
- May not be able to remain when lease term is over. The landlord can, at the end of your lease, invite you to leave.
- The building may be sold, and the new landlord may institute changes to the lease when the previous lease expires.

Renting has fees to be paid at the start of the lease. Typically, when you rent, you will pay first and last months' rent and a security deposit. A security deposit is a sum of money that the landlord holds until the renter leaves the rental property. The deposit will cover repairs for damage to the apartment during the renter's stay but may be returned if the apartment is in good condition. If your landlord runs a credit check on you, the landlord may charge you for that.

Advantages of Buying a Home

The advantages to buying a home mirror the disadvantages of renting, and the disadvantages of home ownership mirror the advantages of renting.

Some advantages to buying a home are:

- There are tax incentives. The interest you pay for your mortgage (more on that later) is deductible on your federal income tax.
- There are no restrictions on pets or occupants, unless laws in your area specify limits for homes.
- You can redecorate any way you wish, limited only by the laws in your area.
- Once your mortgage is set with a fixed-interest rate, your housing cost is fixed.
- Your home grows equity, that is, the difference between what you owe and what the house is worth grows. You can use the equity to secure loans, and you recover the equity (and more if you're fortunate) when you sell the house.
- As long as you pay your mortgage and maintain the home to the standards of your community, you can stay as long as you wish.

Some disadvantages to home ownership are:

- The cost is higher than renting. Mortgages and associated costs are typically higher than rent for a similar living space.
- The owner is responsible for upkeep, maintenance, and repairs. These can be extremely costly.
- The owner cannot walk away from the property. It can be sold, but simply leaving the property, especially if not paid off yet, has serious consequences.

The big question of affordability looms large over the decision to rent or buy. Renting, strictly from an affordability viewpoint, comes with much less initial outlay and smaller commitment. If you do not have sufficient income to regularly save for possibly expensive repairs, or your credit isn't quite as good as it needs to be, then renting may be the best choice. Of course, even if you can afford to buy a home, you may choose to rent based on the comparative advantages.

VIDEO

[Rent or Buy \(https://openstax.org/r/Rent_or_Buy\)](https://openstax.org/r/Rent_or_Buy)

Buying a home really involves two buyers. You and the mortgage company. The mortgage company has interest in the home, as they are providing the funds for the home. They want to protect their investment, and many fees are about the bank as much as the buyer. They fund a mortgage based on the value they assign the property. Not you. This means they will want some certainty that the home is sound, and you are a good investment.

WHO KNEW?

Closing Costs

When a home is bought, there are many costs that need to be paid at the time of purchase, which are lumped under the term closing costs. At the start of 2022, the average closing costs for a single-family home exceeded \$6,800. These costs include:

- The appraisal fee, which is what is paid to someone to establish the home's worth. The value of the home to the bank may differ from what the home is listed for, or what an app tells you the home is worth. It may run approximately \$350.
- The home inspection fee. The inspection should reveal any problems with the house that will need to be fixed either before or after you obtain the home.
- The title search. This is a records search to insure there are no issues with who actually owns the property. It can cost about 0.5% to 1% of the amount you are financing.
- Prepaid taxes. You will need to pay about 6 months of taxes at the time of purchase.
- The credit report fee. This is a fee for checking your credit. You might pay \$25 or more for this.
- The origination fee. This is the price the mortgage company charges you to cover the costs of creating the mortgage. This could be 0.5% to 1% (or more) of the amount you are borrowing.
- The application fee. This is just a processing fee and could come to several hundred dollars.
- The underwriting fee. This covers the cost of verifying your financial qualifications. It could be a flat fee, or some

small percentage of the amount financed. Such as 0.5% or 1%.

- Attorney fees. If you use an attorney, you will have to pay the attorney.
- State of local fees. This may include a filing fee charged by the county or municipality in which you reside.

That's a long list, and it is not even complete. When buying, be prepared to see these costs. It can be surprising. But in the end, you will have equity in the home, which means when you sell your home, you will get some of your money back.

VIDEO

[Closing Costs \(https://openstax.org/r/closing_costs\)](https://openstax.org/r/closing_costs)

In the end, you must weigh your options and carefully consider your priorities in choosing to rent or buy a home.

Mortgages

Some people will purchase a home or condo with cash, but the majority of people will apply for a mortgage. A **mortgage** is a long-term loan and the property itself is the security. The bank decides the minimum down payment (with your input), the payment schedule, the duration of the loan, whether the loan can be assumed by another party, and the penalty for late payments. The title of the home belongs to the bank.

Since a mortgage is a loan, everything about loans from [The Basics of Loans](#) holds true, including the formula for the payments.

Monthly Mortgage Payments

The formula to calculate your monthly payments of principal and interest uses APR as the annual interest rate.

FORMULA

The payment, pmt , per month to pay down a mortgage with beginning principal P is

$pmt = \frac{P \times (r/12) \times (1 + r/12)^{12 \times t}}{(1 + r/12)^{12 \times t} - 1}$, where r is the annual interest rate in decimal form and t is the number of years of the payment.

Note, payment to lenders is always rounded up to the next penny.

To find the total amount of your payments over the life of the loan, multiply your monthly payments by the number of payments.

EXAMPLE 6.110

30-Year Mortgage at 4.8% Interest

Evan buys a house. His 30-year mortgage comes to \$132,650 with 4.8% interest. Find Evan's monthly payments.

✓ Solution

Using the information above, $P = \$132,650$, $r = 0.048$ and $t = 30$. Substituting those values into the formula

$pmt = \frac{P \times (r/12) \times (1 + r/12)^{12 \times t}}{(1 + r/12)^{12 \times t} - 1}$ and calculating, we find the payment is

$$\begin{aligned}
 pmt &= \frac{P \times (r/12) \times (1 + r/12)^{12 \times t}}{(1 + r/12)^{12 \times t} - 1} \\
 &= \frac{\$132,650 \times (0.048/12) \times (1 + 0.048/12)^{12 \times 30}}{(1 + 0.048/12)^{12 \times 30} - 1} \\
 &= \frac{\$132,650 \times (0.004) \times (1.004)^{360}}{(1.004)^{360} - 1} \\
 &= \frac{\$2,233.07781448}{3.20858992551} \\
 &= \$695.97
 \end{aligned}$$

His mortgage payment is \$695.97.

> YOUR TURN 6.110

1. Paulo buys a house. His 20-year mortgage comes to \$153,899 with 4.21% interest. Find Evan's monthly payments.

To find the total amount of your payments over the life of the loan, multiply your monthly payments by the number of payments. This can be useful information, but not too many people reach the end of their mortgage. They tend to move before the mortgage is paid off.

FORMULA

The total paid, T , on an t year mortgage with monthly payments pmt is $T = pmt \times 12 \times t$.

EXAMPLE 6.111

30-Year Mortgage at 5.35% Interest

Cassandra buys a house. Her 30-year mortgage comes to \$99,596 with 5.35% interest. If Cassandra pays off the mortgage over those 30 years, how much will she have paid in total?

✓ Solution

To find the total paid over the life of the mortgage, use the formula $T = pmt \times 12 \times t$. To calculate this, the payment must be found. Using the information above, $P = \$99,596$, $r = 0.0535$ and $t = 30$. Substituting those values into the

formula $pmt = \frac{P \times (r/12) \times (1 + r/12)^{12 \times t}}{(1 + r/12)^{12 \times t} - 1}$ and calculating, we find the payment is

$$\begin{aligned} pmt &= \frac{P \times (r/12) \times (1 + r/12)^{12 \times t}}{(1 + r/12)^{12 \times t} - 1} \\ &= \frac{\$99,596 \times (0.0535/12) \times (1 + 0.0535/12)^{12 \times 30}}{(1 + 0.048/12)^{12 \times 30} - 1} \\ &= \frac{\$99,596 \times (0.004458\bar{3}) \times (1.004458\bar{3})^{360}}{(1.004458\bar{3})^{360} - 1} \\ &= \frac{\$2,202.45911222}{3.96013414693} \\ &= \$556.16 \end{aligned}$$

Using the mortgage payment of \$556.16 and $t = 30$ years in the formula $T = pmt \times 12 \times t$, the total that Cassandra will pay for the mortgage is \$200,217.60.

> YOUR TURN 6.111

1. Arthur buys a house. Their 15-year mortgage comes to \$225,879 with 4.91% interest. If Arthur pays off the mortgage over those 15 years, how much will they have paid in total?

With the principal of the mortgage and how much total is paid over the life of the mortgage, the cost of financing can be found by subtracting the principal of the mortgage from the total paid over the life of the mortgage.

FORMULA

The cost of financing a mortgage, CoF, is $\text{CoF} = T - P$ where P is the mortgage's starting principal and T is the total

paid over the life of the mortgage.

EXAMPLE 6.112

30-Year Mortgage at 5.35% Interest

Cassandra buys a house. Her 30-year mortgage comes to \$99,596 with 5.35% interest. What was Cassandra's cost of financing?

✓ Solution

In [Example 6.111](#), we found that the total Cassandra will pay for the \$99,569 mortgage is \$200,217.60. Subtracting those we find the cost of financing $\text{CoF} = T - P = \$200,217.60 - \$99,596 = \$100,621.60$.

> YOUR TURN 6.112

1. Arthur buys a house. Their 15-year mortgage comes to \$225,879 with 4.91% interest. What was Arthur's cost of financing?

? WHO KNEW?

Private Mortgage Insurance (PMI)

When you purchase a home, you will have to pay a down payment. This means you have money tied to the property, which lenders believe makes you less likely to walk away from a property. The amount of the down payment will be decided between you and the mortgage company. However, if your down payment is less than 20% of the property value, you will be required to pay private mortgage insurance (PMI). This is insurance you pay for so that the mortgage company is protected if *you* default on the loan. It often comes to between 0.5% and 2.25% of the original loan amount. It increases your monthly payment. Once you reach 20% of the loan value, you can request that the PMI be dropped. Even if you do not request cancelling the PMI, it will eventually and automatically be dropped.

For more, see [this article on ways to get rid of PMI \(https://openstax.org/r/mortgage_insurance\)](https://openstax.org/r/mortgage_insurance).

Reading and Interpreting Amortization Tables

Amortization tables were addressed in [The Basics of Loans](#). They are most frequently encountered when analyzing mortgages.

The amortization table for a 30-year mortgage is quite long, containing 360 rows. A full table will not be reproduced here. We can, though, read information from a portion of an amortization table.

EXAMPLE 6.113

Amortization Table for a 30-Year, \$165,900 Mortgage

[Figure 6.27](#) shows a portion of an amortization table for a 30-year, \$165,900 mortgage. Use that table to answer the following questions.

1. What is the interest rate?
2. How much are the payments?
3. How much of payment 175 goes to principal?
4. How much of payment 180 goes to interest?
5. What's the remaining balance on the mortgage after payment 170?

Amortization Schedule Calculator					
Loan Amount		\$165,900.00			
Interest Rate (annual)		5.61%			
Term (Years)		30			
Monthly Payment		\$953.44			
Month	Payment	Principal	Interest	Total Interest	Balance
167	\$953.44	\$385.77	\$567.67	\$114,366.57	\$121,041.52
168	\$953.44	\$387.57	\$565.87	\$114,932.44	\$120,653.95
169	\$953.44	\$389.39	\$564.06	\$115,496.50	\$120,264.56
170	\$953.44	\$391.21	\$562.24	\$116,058.73	\$119,873.35
171	\$953.44	\$393.04	\$560.41	\$116,619.14	\$119,480.32
172	\$953.44	\$394.87	\$558.57	\$117,177.71	\$119,085.44
173	\$953.44	\$396.72	\$556.72	\$117,734.44	\$118,688.73
174	\$953.44	\$398.57	\$554.87	\$118,289.31	\$118,290.15
175	\$953.44	\$400.44	\$553.01	\$118,842.31	\$117,889.71
176	\$953.44	\$402.31	\$551.13	\$119,393.45	\$117,487.41
177	\$953.44	\$404.19	\$549.25	\$119,942.70	\$117,083.22
178	\$953.44	\$406.08	\$547.36	\$120,490.06	\$116,677.14
179	\$953.44	\$407.98	\$545.47	\$121,035.53	\$116,269.16
180	\$953.44	\$409.89	\$543.56	\$121,579.09	\$115,859.27
181	\$953.44	\$411.80	\$541.64	\$122,120.73	\$115,447.47
182	\$953.44	\$413.73	\$539.72	\$122,660.45	\$115,033.75
183	\$953.44	\$415.66	\$537.78	\$123,198.23	\$114,618.09
184	\$953.44	\$417.60	\$535.84	\$123,734.07	\$114,200.48
185	\$953.44	\$419.56	\$533.89	\$124,267.96	\$113,780.93
186	\$953.44	\$421.52	\$531.93	\$124,799.88	\$113,359.41
187	\$953.44	\$423.49	\$529.96	\$125,329.84	\$112,935.92
188	\$953.44	\$425.47	\$527.98	\$125,857.81	\$112,510.45
189	\$953.44	\$427.46	\$525.99	\$126,383.80	\$112,082.99

Figure 6.27 Amortization table

✓ **Solution**

1. Reading at the top of the table, we see the interest rate is 5.61%.
2. Reading from the top of the table or from the column labeled Payment, we see the payments are \$953.44 per month.
3. In the row for payment 175, we see that the amount that goes to principal is \$400.44.
4. In the row for payment 180, we see that the amount that goes to interest is \$543.56.
5. In the row for payment 170, we see the remaining balance is \$119,873.35.

> **YOUR TURN 6.113**

Below is a portion of an amortization table for a 30-year, \$228,320 mortgage. Use that table to answer the following questions.

Amortization Schedule Calculator					
Loan Amount	\$228,320.00				
Interest Rate (annual)	6.10%				
Term (Years)	30				
Monthly Payment	\$1,383.61				
Month	Payment	Principal	Interest	Total Interest	Balance
213	\$1,383.61	\$655.29	\$728.32	\$209,405.96	\$143,013.61
214	\$1,383.61	\$656.61	\$727.00	\$210,130.96	\$142,359.00
215	\$1,383.61	\$659.95	\$723.66	\$210,854.62	\$141,699.05
216	\$1,383.61	\$663.30	\$720.30	\$211,574.92	\$141,035.74
217	\$1,383.61	\$666.68	\$716.93	\$212,291.85	\$140,369.07
218	\$1,383.61	\$670.06	\$713.54	\$213,005.39	\$139,699.00
219	\$1,383.61	\$673.47	\$710.14	\$213,715.53	\$139,025.53
220	\$1,383.61	\$676.89	\$706.71	\$214,422.24	\$138,348.64
221	\$1,383.61	\$680.34	\$703.27	\$215,125.52	\$137,668.30
222	\$1,383.61	\$683.79	\$699.81	\$215,825.33	\$136,984.51
223	\$1,383.61	\$687.27	\$696.34	\$216,521.67	\$136,297.24
224	\$1,383.61	\$690.76	\$692.84	\$217,214.51	\$135,606.48
225	\$1,383.61	\$694.27	\$689.33	\$217,903.84	\$134,912.20
226	\$1,383.61	\$697.80	\$685.80	\$218,589.65	\$134,214.40
227	\$1,383.61	\$701.35	\$682.26	\$219,271.90	\$133,513.05
228	\$1,383.61	\$704.92	\$678.69	\$219,950.60	\$132,808.13
229	\$1,383.61	\$708.50	\$675.11	\$220,625.70	\$132,099.63
230	\$1,383.61	\$712.10	\$671.51	\$221,297.21	\$131,387.53
231	\$1,383.61	\$715.72	\$667.89	\$221,965.10	\$130,671.81
232	\$1,383.61	\$719.36	\$664.25	\$222,629.35	\$129,952.45
233	\$1,383.61	\$723.02	\$660.59	\$223,289.94	\$129,229.44
234	\$1,383.61	\$726.69	\$656.92	\$223,946.85	\$128,502.75
235	\$1,383.61	\$730.38	\$653.22	\$224,600.08	\$127,772.36
236	\$1,383.61	\$734.10	\$649.51	\$225,249.59	\$127,038.26

Amortization table

1. What is the interest rate?
2. How much are the payments?
3. How much of payment 235 goes to principal?
4. How much of payment 215 goes to interest?
5. What's the remaining balance on the mortgage after payment 227?
6. At what payment does the amount that is applied to mortgage finally exceed the amount applied to interest?

Escrow Payments

The last few examples have looked at mortgage payments, which cover the principal and interest. However, when you take out a mortgage, the payment is sometimes much higher than that. This is because your mortgage company also has you pay into an **escrow account**, which is a savings account maintained by the mortgage company.

Your insurance payments will be set by your insurer and the mortgage company will pay them on time for you from your escrow account. Your property taxes are set by where you live and are typically a percentage of your property's **assessed value**. The assessed value is the estimation of the value of your home and does not necessarily reflect the purchase or resale value of the home. Your property taxes will also be paid on time by the mortgage company from your escrow account.

For example, in Kalamazoo, Michigan, the effective tax rate for property is 1.69% of the assessed value of the home. These escrow payments, which cover bills for the home, can increase the monthly payments for your home well beyond the basic principal and interest payment.

EXAMPLE 6.114

Adding Escrow Payments to Mortgage Payments

Jenna decides to purchase a home, with mortgage of \$108,450 at 6% interest for 30 years. The assessed value of her home is \$75,600. Her property taxes come to 5.7% of her assessed value. Jenna also has to pay her home insurance every 6 months, which is \$744 per six months. How much, including escrow, will Jenna pay per month?

Solution

Using the payment function to find her mortgage payments, $pmt = \frac{P \times (r/12) \times (1 + r/12)^{12 \times t}}{(1 + r/12)^{12 \times t} - 1}$, with $P = \$108,405$, $r =$

0.06, and $t = 30$, her payments are

$$\begin{aligned} pmt &= \frac{P \times (r/12) \times (1 + r/12)^{12 \times t}}{(1 + r/12)^{12 \times t} - 1} \\ &= \frac{\$108,405 \times (0.06/12) \times (1 + 0.06/12)^{12 \times 30}}{(1 + 0.06/12)^{12 \times 30} - 1} \\ &= \$649.95 \end{aligned}$$

Jenna also pays into escrow 1/12 of her property taxes per month. Her property taxes are 5.7% of the assessed value of \$75,600, which comes to $0.057 \times \$75,600 = \$4,309.20$. This is an annual tax, so she pays 1/12 of that each month, or \$359.10. Jenna's home insurance is \$744 per 6 months, so each month she pays \$124.00 for insurance. Adding these together, her monthly payment is $\$649.95 + \$359.10 + \$124.00 = \$1,133.05$. This is quite a bit more than the \$649.95 for the principal and interest.

> YOUR TURN 6.114

1. Destiny decides to purchase a home, with mortgage of \$159,195.50 at 5.75% interest for 30 years. The assessed value of her home is \$100,000. Her property taxes come to 5.42% of her assessed value. Destiny also has to pay her home insurance every 6 months, and that comes to \$843 per 6 months. How much, including escrow, will Destiny pay per month?

Check Your Understanding

78. Does renting or buying have tax advantages?
79. Which has more restrictions, renting or buying?
80. Which has housing cost that does not change?
81. What is the name given to a loan for a home?
82. What are the monthly payments for a 30-year mortgage of \$108,993 with 6.14% interest.
83. What is the cost of financing for a 30-year mortgage of \$108,993 with 6.14% interest if the mortgage is paid off?
84. Consider the following amortization table. What is the amount of payment 141 that goes to principal?

Amortization Schedule Calculator					
Loan Amount	\$274,800.00				
Interest Rate (annual)	5.90%				
Term (Years)	30				
Monthly Payment	\$1,629.94				
Month	Payment	Principal	Interest	Total Interest	Balance
136	\$1,629.94	\$540.64	\$1,089.30	\$167,882.89	\$221,011.17
137	\$1,629.94	\$543.30	\$1,086.64	\$168,969.53	\$220,467.87
138	\$1,629.94	\$545.97	\$1,083.97	\$170,053.50	\$219,921.90
139	\$1,629.94	\$548.66	\$1,081.28	\$171,134.78	\$219,373.24
140	\$1,629.94	\$551.35	\$1,078.59	\$172,213.37	\$218,821.89
141	\$1,629.94	\$554.06	\$1,075.87	\$173,289.24	\$218,267.83
142	\$1,629.94	\$556.79	\$1,073.15	\$174,362.39	\$217,711.04
143	\$1,629.94	\$559.53	\$1,070.41	\$175,432.80	\$217,151.51
144	\$1,629.94	\$562.28	\$1,067.66	\$176,500.47	\$216,589.23
145	\$1,629.94	\$565.04	\$1,064.90	\$177,565.36	\$216,024.19
146	\$1,629.94	\$567.82	\$1,062.12	\$178,627.48	\$215,456.37
147	\$1,629.94	\$570.61	\$1,059.33	\$179,686.81	\$214,885.76
148	\$1,629.94	\$573.42	\$1,056.52	\$180,743.33	\$214,312.34
149	\$1,629.94	\$576.24	\$1,053.70	\$181,797.03	\$213,736.10
150	\$1,629.94	\$579.07	\$1,050.87	\$182,847.90	\$213,157.03
151	\$1,629.94	\$581.92	\$1,048.02	\$183,895.92	\$212,575.12
152	\$1,629.94	\$584.78	\$1,045.16	\$184,941.09	\$211,990.34
153	\$1,629.94	\$587.65	\$1,042.29	\$185,983.37	\$211,402.69
154	\$1,629.94	\$590.54	\$1,039.40	\$187,022.77	\$210,812.14
155	\$1,629.94	\$593.45	\$1,036.49	\$188,059.26	\$210,218.70
156	\$1,629.94	\$596.36	\$1,033.58	\$189,092.84	\$209,622.33
157	\$1,629.94	\$599.30	\$1,030.64	\$190,123.48	\$209,023.04
158	\$1,629.94	\$602.24	\$1,027.70	\$191,151.18	\$208,420.80

85. What is the name of the account that the mortgage company holds your taxes and insurance in?



SECTION 6.12 EXERCISES

In the following exercises, indicate if the advantage listed is for renting or buying a home.

1. Short-term commitment.
2. Tax advantage.
3. Freedom to remodel.
4. Builds equity.
5. Cost is lower.
6. You do not pay for repairs.
7. No pet restrictions
8. More flexibility to move.
9. Housing cost is fixed.
10. May have other amenities.

In the following exercises, find the mortgage payment for the given loan amount, interest rate, and term.

11. Loan amount is \$78,560, interest rate is 5.87%, 30-year mortgage.
12. Loan amount is \$125,800, interest rate is 6.5%, 30-year mortgage.
13. Loan amount is \$96,400, interest rate is 4.9%, 15-year mortgage.
14. Loan amount is \$267,450, interest rate is 5.25%, 20-year mortgage.

In the following exercises, find the total paid on the mortgage if it is fully paid through the term.

15. Loan amount is \$78,560, interest rate is 5.87%, 30-year mortgage.
16. Loan amount is \$125,800, interest rate is 6.5%, 30-year mortgage.
17. Loan amount is \$96,400, interest rate is 4.9%, 15-year mortgage.
18. Loan amount is \$267,450, interest rate is 5.25%, 20-year mortgage.

In the following exercises, find the cost of financing for the mortgages if they are fully paid.

19. Loan amount is \$78,560, interest rate is 5.87%, 30-year mortgage.
20. Loan amount is \$125,800, interest rate is 6.5%, 30-year mortgage.
21. Loan amount is \$96,400, interest rate is 4.9%, 15-year mortgage.
22. Loan amount is \$267,450, interest rate is 5.25%, 20-year mortgage.

In the following exercises, use the amortization table to answer the question.

Amortization Schedule Calculator					
Loan Amount	\$94,176.00				
Interest Rate (annual)	6.30%				
Term (Years)	30				
Monthly Payment	\$582.92				
Month	Payment	Principal	Interest	Total Interest	Balance
147	\$582.92	\$190.09	\$392.83	\$66,149.28	\$74,635.48
148	\$582.92	\$191.09	\$391.84	\$66,541.12	\$74,444.39
149	\$582.92	\$192.09	\$390.83	\$66,931.95	\$74,252.30
150	\$582.92	\$193.10	\$389.82	\$67,321.77	\$74,059.20
151	\$582.92	\$194.11	\$388.81	\$67,710.59	\$73,865.09
152	\$582.92	\$195.13	\$387.79	\$68,098.38	\$73,669.96
153	\$582.92	\$196.16	\$386.77	\$68,485.14	\$73,473.80
154	\$582.92	\$197.19	\$385.74	\$68,870.88	\$73,276.61
155	\$582.92	\$198.22	\$384.70	\$69,255.58	\$73,078.39
156	\$582.92	\$199.26	\$383.66	\$69,639.25	\$72,879.13
157	\$582.92	\$200.31	\$382.62	\$70,021.86	\$72,678.82
158	\$582.92	\$201.36	\$381.56	\$70,403.42	\$72,477.46
159	\$582.92	\$202.42	\$380.51	\$70,783.93	\$72,275.05
160	\$582.92	\$203.48	\$379.44	\$71,163.38	\$72,071.57
161	\$582.92	\$204.55	\$378.38	\$71,541.75	\$71,867.02
162	\$582.92	\$205.62	\$377.30	\$71,919.05	\$71,661.40
163	\$582.92	\$206.70	\$376.22	\$72,295.28	\$71,454.69
164	\$582.92	\$207.79	\$375.14	\$72,670.41	\$71,246.91
165	\$582.92	\$208.88	\$374.05	\$73,044.46	\$71,038.03
166	\$582.92	\$209.97	\$372.95	\$73,417.41	\$70,828.06
167	\$582.92	\$211.08	\$371.85	\$73,789.26	\$70,616.98
168	\$582.92	\$212.18	\$370.74	\$74,159.99	\$70,404.79
169	\$582.92	\$213.30	\$369.63	\$74,529.62	\$70,191.50

23. What is the term of the mortgage?
24. How much of payment 165 applies to interest?

25. What is the remaining balance after payment 155?
 26. How much total interest was paid after payment 149?

In the following exercises, use the amortization schedule to answer the question.

Amortization Schedule Calculator					
Loan Amount	\$168,458.00				
Interest Rate (annual)	5.85%				
Term (Years)	15				
Monthly Payment	\$1,407.93				
Month	Payment	Principal	Interest	Total Interest	Balance
93	\$1,407.93	\$917.74	\$490.19	\$62,112.60	\$99,633.22
94	\$1,407.93	\$922.22	\$485.71	\$62,598.32	\$98,711.00
95	\$1,407.93	\$926.71	\$481.22	\$63,079.53	\$97,784.29
96	\$1,407.93	\$931.23	\$476.70	\$63,556.23	\$96,853.06
97	\$1,407.93	\$935.77	\$472.16	\$64,028.39	\$95,917.29
98	\$1,407.93	\$940.33	\$467.60	\$64,495.99	\$94,976.96
99	\$1,407.93	\$944.92	\$463.01	\$64,959.00	\$94,032.04
100	\$1,407.93	\$949.52	\$458.41	\$65,417.40	\$93,082.52
101	\$1,407.93	\$954.15	\$453.78	\$65,871.18	\$92,128.37
102	\$1,407.93	\$958.80	\$449.13	\$66,320.31	\$91,169.56
103	\$1,407.93	\$963.48	\$444.45	\$66,764.76	\$90,206.09
104	\$1,407.93	\$968.17	\$439.75	\$67,204.51	\$89,237.91
105	\$1,407.93	\$972.89	\$435.03	\$67,639.55	\$88,265.02
106	\$1,407.93	\$977.64	\$430.29	\$68,069.84	\$87,287.38
107	\$1,407.93	\$982.40	\$425.53	\$68,495.37	\$86,304.98
108	\$1,407.93	\$987.19	\$420.74	\$68,916.10	\$85,317.79
109	\$1,407.93	\$992.00	\$415.92	\$69,332.03	\$84,325.78
110	\$1,407.93	\$996.84	\$411.09	\$69,743.12	\$83,328.94
111	\$1,407.93	\$1,001.70	\$406.23	\$70,149.34	\$82,327.24
112	\$1,407.93	\$1,006.58	\$401.35	\$70,550.69	\$81,320.66
113	\$1,407.93	\$1,011.49	\$396.44	\$70,947.13	\$80,309.17
114	\$1,407.93	\$1,016.42	\$391.51	\$71,338.63	\$79,292.74
115	\$1,407.93	\$1,021.38	\$386.55	\$71,725.19	\$78,271.37

27. What is the interest rate for the mortgage?
 28. How much of payment 110 applies to principal?
 29. What is the remaining balance after payment 94?
 30. How much total interest was paid after payment 111?

In the following exercises, find the total monthly payment including both the mortgage payment and the escrow payment.

31. Mortgage of \$87,690 at 6.2% interest for 30 years. Assessed value of the home is \$75,600. Property taxes come to 5.65% of assessed value. Home insurance of \$815 paid every 6 months.
 32. Mortgage of \$143,900 at 5.05% interest for 30 years. Assessed value of the home is \$90,150. Property taxes come to 5.88% of assessed value. Home insurance of \$924 paid every 6 months.
 33. Mortgage of \$65,175 at 6.48% interest for 30 years. Assessed value of the home is \$62,800. Property taxes come to 6.75% of assessed value. Home insurance of \$558 paid every 6 months.
 34. Mortgage of \$245,950 at 5.35% interest for 30 years. Assessed value of the home is \$156,500. Property taxes come to 6.41% of assessed value. Home insurance of \$972 paid every 6 months.

For the following exercises, read the following: **Fifteen-year mortgage compared to 30-year mortgage.** Mortgage interest rates are often higher for 30-year mortgages than 15-year mortgages. However, the payments for 15-year mortgages are considerably higher. The following exercises explore the difference between a 15- and 30-year mortgage for a mortgage of \$100,000.

35. The 15-year mortgage interest rate is 5.65%.
 a. Find the payment.
 b. Determine the total that would be paid if the mortgage was completed.
 c. Find the cost of financing for this mortgage.
 36. The 30-year mortgage rate is 6.4%.
 a. Find the payment.
 b. Determine the total that would be paid if the mortgage was completed.
 c. Find the cost of financing for this mortgage.
 37. How different are the payments, the total paid, and the cost to finance?
 38. Summarize the answer in the previous question.

For the following exercises, read the following: **Fifteen-year mortgage compared to 30-year mortgage.** A 15-year